

PART A — Theory & Definitions

Introduction

Statistics plays an important role in understanding, organizing, analyzing, and interpreting data in real-world situations. In socio-economic research, statistical techniques help researchers understand household income patterns, demographic characteristics, educational status, and economic conditions. In this project, statistical concepts are applied to a household dataset containing information such as age, income, family size, education level, house ownership, and area type.

The purpose of this theoretical section is to explain important statistical concepts including types of data, descriptive and inferential statistics, measures of central tendency, measures of dispersion, probability distributions, skewness, kurtosis, percentiles, quartiles, and five-number summary.

1. Types of Data

Data refers to raw facts, values, or observations collected for analysis. In statistics, data is mainly divided into two categories:

1. Numerical Data
 2. Categorical Data
-

1.1 Numerical Data

Numerical data consists of measurable quantities represented using numbers. Mathematical operations such as addition, subtraction, multiplication, division, average, variance, and standard deviation can be performed on numerical data.

Numerical data is further divided into:

- Discrete Data
- Continuous Data

Discrete Numerical Data

Discrete data contains countable values.

Example

- Family size

- Number of household members

Continuous Numerical Data

Continuous data contains measurable values that can take any value within a range.

Example

- Household income
- Age of household head

Examples from Household Dataset

Column Name	Type
Age_of_Household_Head	Numerical
Household_Income	Numerical
Family_Size	Numerical

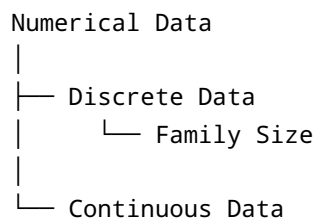
Example Explanation

If a household has:

- Age = 45
- Income = 75,000
- Family Size = 5

These values are numerical because they can be measured and mathematical operations can be applied to them.

Diagram — Numerical Data



└─ Age
└─ Household Income

1.2 Categorical Data

Categorical data represents labels, names, groups, or categories rather than measurable numeric quantities. Arithmetic operations cannot be meaningfully performed on categorical data.

Categorical data helps classify observations into different groups.

Examples from Household Dataset

Column Name	Categories
Education_Level	Primary, Secondary, Graduate, Post-Grad
Owns_House	Yes / No
Urban_Rural	Urban / Rural
Household_ID	H001, H002, H003

Example Explanation

A household may belong to:

- Education Level = Graduate
- Owns House = Yes
- Area = Urban

These values represent categories rather than measurable quantities.

Diagram — Categorical Data

```
Categorical Data
|
└─ Education Level
   |
   └─ Primary
   |
   └─ Secondary
   |
   └─ Graduate
```



2. Types of Statistics

Statistics is the science of collecting, organizing, analyzing, interpreting, and presenting data. Statistics is mainly divided into two categories:

- 1. Descriptive Statistics
- 2. Inferential Statistics

2.1 Descriptive Statistics

Descriptive statistics summarizes and describes the important features of a dataset. It helps in understanding the overall structure and behavior of data.

Descriptive statistics includes:

- Mean
- Median
- Mode
- Range
- Variance
- Standard Deviation
- Quartiles
- Percentiles

Examples of Descriptive Statistics

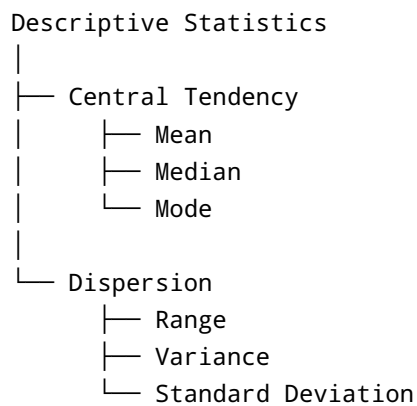
Statistical Measure	Purpose
Mean	Average value

Statistical Measure	Purpose
Median	Middle value
Mode	Most frequent value
Standard Deviation	Measures spread of data
Variance	Measures variability

Example from Dataset

The average household income can be calculated using the mean. The spread of income values can be measured using standard deviation.

Diagram — Descriptive Statistics



2.2 Inferential Statistics

Inferential statistics uses sample data to make predictions, conclusions, or decisions about a larger population. It helps researchers estimate future trends and relationships.

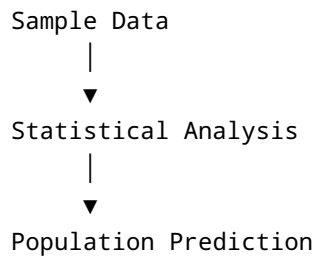
Inferential statistics is commonly used in:

- Forecasting
- Prediction
- Hypothesis testing
- Decision making

Example

Using household income data from 500 households to predict income trends for an entire city population.

Diagram — Inferential Statistics



3. Measures of Central Tendency

Measures of central tendency describe the center or typical value of a dataset.

The three major measures are:

1. Mean
2. Median
3. Mode

3.1 Mean

The mean is the arithmetic average of all observations in a dataset.

It is calculated by dividing the sum of all values by the total number of observations.

Formula

Mean Formula:

$$\bar{x} = (\sum x) / n$$

Where:

- Σx = Sum of observations
- n = Total number of observations

Example

Income values:

20,000 30,000 40,000

Mean = $(20,000 + 30,000 + 40,000) / 3$

Mean = 30,000

Diagram — Mean

20k ----- 30k ----- 40k
 ↑
 Mean

3.2 Median

The median is the middle value of a dataset after arranging values in ascending or descending order.

If the dataset contains an even number of observations, the median is calculated as the average of the two middle values.

Example

20, 30, 40, 50, 60

Median = 40

Diagram — Median

20 30 [40] 50 60
↑
Median

3.3 Mode

The mode is the value that occurs most frequently in a dataset.

A dataset may have:

- One mode
- Multiple modes
- No mode

Example

20, 30, 30, 40, 50

Mode = 30

Diagram — Mode

20 30 30 40 50
↑
Mode

Difference Between Mean, Median, and Mode

Measure	Meaning
Mean	Arithmetic average
Median	Middle value

Measure	Meaning
Mode	Most repeated value

4. Measures of Dispersion

Measures of dispersion describe how spread out the data values are from the center.

Major measures include:

- Range
- Variance
- Standard Deviation
- Interquartile Range

4.1 Range

Range is the difference between the maximum and minimum values.

Formula

Range = Maximum Value – Minimum Value

Example

Maximum Income = 150,000 Minimum Income = 20,000

Range = 130,000

Diagram — Range

20k ----- 150k
 |<----- Range ----->|

4.2 Variance

Variance measures the average squared deviation of observations from the mean.

A higher variance indicates greater spread in the data.

Formula

$$\sigma^2 = \Sigma(x - \mu)^2 / N$$

Where:

- x = observation
- μ = mean
- N = total observations

4.3 Standard Deviation

Standard deviation is the square root of variance.

It measures how far values are spread from the mean.

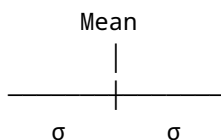
A small standard deviation indicates data values are close to the mean.

A large standard deviation indicates wide variation.

Formula

$$\sigma = \sqrt{[\Sigma(x - \mu)^2 / N]}$$

Diagram — Standard Deviation



5. Gaussian Distribution

Gaussian Distribution, also called Normal Distribution, is a symmetric bell-shaped probability distribution.

In a perfect normal distribution:

- Mean = Median = Mode

The majority of observations are concentrated around the center.

Characteristics of Gaussian Distribution

- Symmetric shape
 - Bell-shaped curve
 - Equal mean, median, and mode
 - Most values cluster near the center
-

Formula

$$f(x) = [1 / (\sigma\sqrt{2\pi})] e^{[-(x-\mu)^2 / 2\sigma^2]}$$

Where:

- μ = mean
 - σ = standard deviation
-

Diagram — Gaussian Distribution



Example

Household income distribution may approximately follow a bell-shaped pattern.

6. Log Normal Distribution

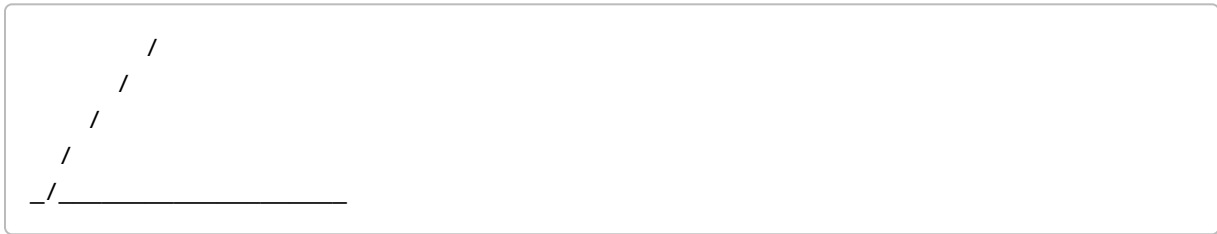
A Log Normal Distribution occurs when the logarithm of a variable follows a normal distribution.

This distribution is commonly used for:

- Income
- Population growth
- Financial data

In socio-economic data, income often follows a log-normal distribution because a few households earn extremely high incomes.

Diagram — Log Normal Distribution



7. 3-Sigma Rule

The 3-Sigma Rule describes how data is distributed around the mean in a normal distribution.

According to this rule:

- 68% data lies within 1 standard deviation
- 95% data lies within 2 standard deviations
- 99.7% data lies within 3 standard deviations

Formula

$$\mu \pm 3\sigma$$

Diagram — 3-Sigma Rule

|----|----|----| Mean |----|----|----|
-3 σ -2 σ -1 σ +1 σ +2 σ +3 σ

8. Percentiles

Percentiles divide data into 100 equal parts.

Percentiles help identify the relative position of a value within a dataset.

Example

If a household income is in the 90th percentile, it means the income is higher than 90% of households.

Diagram — Percentiles

0% ---- 25% ---- 50% ---- 75% ---- 100%

9. Quartiles

Quartiles divide data into four equal parts.

Quartile	Meaning
Q1	First 25% of data
Q2	Median (50%)
Q3	First 75% of data

Diagram — Quartiles

Min --- Q1 --- Q2 --- Q3 --- Max

Example

If $Q1 = 30,000$ and $Q3 = 80,000$:

Then the middle 50% of income values lie between 30,000 and 80,000.

10. Five Number Summary

The five-number summary is used to describe the distribution of data.

It includes:

1. Minimum
2. First Quartile (Q1)
3. Median (Q2)
4. Third Quartile (Q3)
5. Maximum

Diagram — Five Number Summary

Min --- Q1 --- Median --- Q3 --- Max

11. Skewness

Skewness measures the asymmetry of a distribution.

Types of Skewness

Positive Skewness

The tail extends toward the right side.

This occurs when a few values are extremely large.

Negative Skewness

The tail extends toward the left side.

Zero Skewness

Distribution is symmetric.

Formula

$$\text{Skewness} = [\Sigma(x_i - \bar{x})^3 / n] / s^3$$

Diagram — Positive Skewness

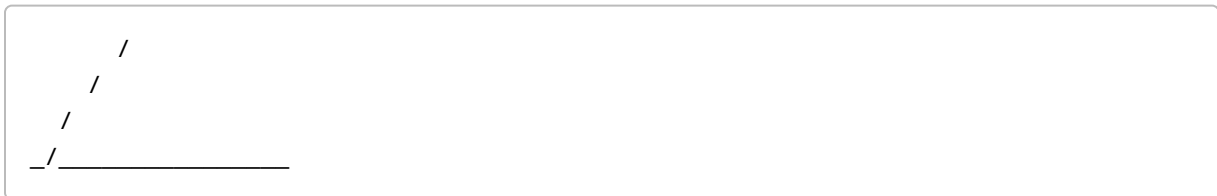


Diagram — Negative Skewness



Example

Household income data is often positively skewed because a small number of households earn very high income.

12. Kurtosis

Kurtosis measures the peakedness or flatness of a distribution.

Types of Kurtosis

Leptokurtic

- Sharp peak
- Heavy tails
- More outliers

Platykurtic

- Flat peak
- Fewer outliers

Mesokurtic

- Normal distribution shape
-

Formula

$$\text{Kurtosis} = [\sum(x_i - \bar{x})^4 / n] / s^4$$

Diagram — Kurtosis

Leptokurtic



Mesokurtic



Platykurtic



Example

High kurtosis in household income indicates the presence of outliers and extreme income values.

Conclusion

The theoretical concepts discussed in this section form the foundation of statistical data analysis. Understanding data types, measures of central tendency, dispersion, distributions, skewness, kurtosis, percentiles, and quartiles helps in analyzing socio-economic household data effectively. These concepts are essential for interpreting real-world data patterns and drawing meaningful conclusions from statistical analysis.